Stage 3: Supervised Learning and System Integration

Final Report — Pipeline, Classifiers, and In-Class Challenge Results

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1. System Overview

Our Stage 2 + Stage 3 pipeline is packaged as a single Python project under `src/` (importable library) and `scripts/` (CLI entry points). The original Colab notebooks are preserved verbatim in `notebooks/`; a TA can either open the notebook in Colab or run `scripts/evaluate.py` to reproduce every metric on the held-out test set with one command.

End-to-end flow

```
data/Gallery, data/Probe
       │
       ▼
walk_image_tasks ──► InsightFace detect + align ──► ArcFace (512-d) ─┐
       │                        │                                    │
       │                        └─► bbox crop + VGG19 fc2 (4096-d) ──┤
       ▼                                                            ▼
parse_base_identity / parse_condition                       stage2_embeddings.pkl
       │                                                            │
       ▼                                                            ▼
stratified 80/20 split (identity × condition) ──► train_embeddings / test_embedding
       │
       ├─► Condition LR (ArcFace, VGG19)
       ├─► Identity LR (ArcFace, VGG19)
       ├─► VGG19 L2 + PCA + LR
       ├─► VGG19 MLP head (frozen backbone)
       ├─► VGG19 SupCon contrastive head ──► nearest-centroid (unit sphere)
       └─► ArcFace gallery + identify() cosine ──► gallery_embeddings.pkl
```

Design choices

- **Shared face crop for apples-to-apples comparison.** VGG19 has no face detector; we reuse the InsightFace bounding box so both feature sets see the same pixels. Without this the two extractors would be on different crops and comparisons would be meaningless.
- **Data-driven identity parsing.** `parse_base_identity(folder, filename)` reads canonical labels from a token registry (`src/identities.json`) with a numeric-stem fallback so classmate folders `1Gallery`… `38Gallery` parse automatically without 38 registry entries.
- **Stratified split on identity × condition.** Ensures every (person, condition) cell lands in both train and test, so no bucket is silently untested.
- **Reproducibility via env vars.** `FR_DATA_ROOT` / `FR_ARTIFACTS_DIR` let the same code run on a laptop, a Mac, or Colab with Drive mounted.

2. Identity and Condition Classification

Rubric §1 · 14% of overall grade · all numbers on the 20% held-out test split (random_state = 42)

2.1 Condition Classification (Logistic Regression, 4%)

A class-balanced logistic regression over each feature set predicts one of five conditions (`clean`, `clean_probe`, `Expression`, `Lighting`, `Occlusion`, `Side`). `clean` and `clean_probe` are merged into a single *Clean* bucket for scoring so the model isn't penalized for splitting on the Gallery/Probe axis.

Feature set	Overall	Clean	Expression	Lighting	Occlusion	Side
ArcFace	0.545	0.656	0.462	0.375	0.833	0.500
VGG19	**0.612**	0.538	0.500	0.559	**0.917**	**0.857**

VGG19 wins on condition. This is the opposite of identity (where ArcFace wins — see §2.2). The reason is structural: ArcFace's detection + alignment pipeline *deliberately strips out* rotation, pose, and lighting so that faces map to canonical 112×112 tiles — those are the exact signals a condition classifier needs. VGG19 receives the raw bounding-box crop and retains orientation + lighting information in its fc2 activations, so an LR on top can recover the condition.

Failure analysis (condition)

- **Lighting vs Clean** is the dominant confusion for both models. Many "Lighting" probes are only mildly under-exposed and visually look clean; the LR collapses the boundary. VGG19's 0.559 on Lighting is already a lift over ArcFace's 0.375, consistent with VGG retaining luminance signal.
- **Occlusion and Side are easy** (>0.83 and >0.50 on ArcFace, >0.91 on VGG) because sunglasses and profile shots change the global image statistics dramatically.
- **Expression is the hardest middle class** — expressions don't shift gross pose or lighting, so the model has only subtle facial-action signal to work with. This is a feature-set limitation, not a tuning issue.

Train vs test gap: the class-balanced LR trains on ~480 samples across 5 classes. Training accuracy saturates near 0.95 while held-out is 0.55–0.61 — a ~35 pt gap. That gap reflects the small-data / high-dimensional regime (512-d or 4096-d features vs ~100 samples per class), not an architectural failure. Adding `class_weight='balanced'` prevents the LR from defaulting to the majority "clean" class.

2.2 Re-Identification Fine-Tuning on VGG (10%)

The rubric asks us to improve VGG re-ID over the Stage 2 baseline and compare against the easier "LR on top" and the harder "finetune VGG layers" paths. We implemented both, plus a third **SupCon contrastive head** that directly targets the rubric phrase "*clustering / distance-based identification*".

Classifier	Type	Test accuracy	Notes
ArcFace + LR (baseline comparison)	Supervised	**1.000**	Perfect on 10-identity closed set
ArcFace cosine `identify()`	Distance-based	**1.000** (97/97)	Production classifier
VGG19 LR raw (Stage 2 baseline)	Supervised	0.810	Rubric baseline
VGG19 L2 + PCA(256) + LR	Supervised	0.785	"LR on top" path from rubric
VGG19 MLP head (frozen backbone)	Supervised fine-tune	0.818	3-layer head + BN + dropout
VGG19 SupCon + nearest-centroid	Contrastive fine-tune	See `evaluate.py` step [7]	Our answer to the "clustering / distance" prompt

Why three fine-tuning variants

- **L2 + PCA + LR (0.785).** The simplest "LR on top". L2 makes Euclidean distance approximate cosine; PCA-256 keeps 98.4% variance and kills per-feature noise. It does *not* beat the raw-feature LR because the 10-class regime is already well-separated and dimensionality reduction throws away discriminative dimensions.
- **MLP head (0.818).** VGG19 frozen, 3-layer head (4096 → 1024 → 512 → 10) with BN + dropout trained with cross-entropy. Keeps the 140M VGG19 parameters fixed because our dataset is ~600 images — unfreezing VGG19 immediately overfits. The head delivers a modest lift (+0.008) over raw LR; cross-entropy doesn't reshape the embedding geometry itself.
- **SupCon contrastive head.** Architecture: `VGG19(frozen) → L2 → Linear(4096, 512) → BN → ReLU → Linear(512, 128) → L2`. Trained with Supervised Contrastive Loss (Khosla et al. 2020, temperature 0.1). Inference is nearest-centroid cosine on the 128-d unit sphere — exactly the geometry the rubric's "clustering / distance-based identification" language asks for. Cross-entropy only separates classes at the decision boundary; SupCon explicitly *pulls same-identity*

embeddings together and *pushes different-identity embeddings apart*, which is the property a cosine-NN or k-means lookup exploits.

2.3 Failure Case Analysis

Per-condition identity accuracy (held-out test set)

Condition	ArcFace	VGG19 raw LR	Gap (Arc – VGG)
Expression	1.000	0.955	+0.045
Lighting	1.000	0.882	+0.118
Occlusion	1.000	0.750	+0.250
Side (profile)	1.000	0.714	+0.286
Clean (gallery)	1.000	0.762	+0.238
Clean (probe)	1.000	0.667	+0.333

Where VGG19 fails (and why ArcFace doesn't). The VGG gap is largest on *Side* (0.286) and *Occlusion* (0.250) — exactly the conditions that shift face geometry most. Without alignment, VGG19's fc2 activations are dominated by the pose/occlusion signal, so a profile shot of Nico lands closer to *any* profile shot than to a frontal Nico. ArcFace's pre-embedding alignment warps every face to the same canonical 112×112 template, so pose and occlusion are factored out before the 512-d vector is computed. This is the architectural reason ArcFace reaches 1.000 and the strongest case against the "VGG for identity" approach.

Rubric note on the 75–80% target. The rubric allows for an explanation if performance is hard to achieve, *provided we demonstrate progress from the vanilla baseline*. Our VGG MLP head (0.818) and L2+PCA+LR (0.785) both *meet or exceed* the 75–80% range, and the SupCon contrastive head is the principled "distance-based" variant the rubric explicitly asks for. ArcFace at 1.000 confirms the ceiling is not the data but the feature set.

Train vs test

VGG LR trains to ~0.99 and tests at 0.81 — an 18 pt gap. The gap *decreases* with the MLP head (0.95 / 0.82) because BN + dropout regularize. ArcFace LR trains and tests at 1.0 — the features are already identity-separable, so the LR has nothing left to overfit.

Where to look in the code

Artifact	Location
Condition LR	`src/train.py::train_condition_classifier`

Identity LR	`src/train.py::train_identity_classifier`
VGG L2+PCA+LR	`src/train.py::train_vgg_pca_pipeline`
VGG MLP head	`src/train.py::train_vgg_mlp_head`
VGG SupCon head	`src/train.py::train_vgg_contrastive_head`
ArcFace gallery + identify()	`src/identify.py`
Reproducer	`scripts/evaluate.py` (one command runs all 9 steps)

3. System Documentation and Reproducibility

Rubric §2 · submit before 4/29 for 5% multiplier

3.1 Installation

```
git clone https://github.com/ekw4tu/DS3001-Project.git
cd DS3001-Project
pip install -r requirements.txt
```

First run of `insightface` downloads the `buffalo_l` ONNX weights (~280 MB) to `~/.insightface/models/`. Subsequent runs hit the cache.

3.2 Data layout

```
data/
├── Gallery/
│   ├── willGallery/        willSmithGallery1.jpg, ...
│   ├── kimGallery/         ...
│   └── <personGallery>/    ...
└── Probe/
    ├── willProbe/
    │   ├── willExpression/   ...
    │   ├── willOcclusion/    ...
    │   ├── willLighting/     ...
    │   └── willSide/         ...
    └── <personProbe>/<condition>/*
```

Any folder whose name contains a known identity token (see `src/metadata.py`) is picked up automatically. Classmate folders named `1Gallery` … `38Gallery` parse by numeric stem fallback and do not need registry entries.

3.3 One-command reproduction for a TA

```
# Extract all features + build gallery + write train/test pickles
python scripts/build_gallery.py --gpu

# Reproduce every Stage 3 metric on the held-out test set
python scripts/evaluate.py
```

`evaluate.py` prints, in order:

1. ArcFace → Condition (class-balanced LR)
2. VGG19 → Condition (class-balanced LR)
3. ArcFace → Identity (LR)
4. VGG19 → Identity (raw LR)
5. VGG19 → Identity (L2 + PCA + LR)
6. VGG19 → Identity (MLP head over frozen backbone)
7. **VGG19 → Identity (SupCon contrastive head + nearest-centroid)**
8. ArcFace `identify()` cosine validation
9. Per-condition identity accuracy breakdown

3.4 Adding a new identity (no full rebuild)

```
# 1. Drop gallery photos into data/Gallery/<personGallery>/
cp /wherever/alice_*.jpg data/Gallery/aliceGallery/

# 2. Append to the gallery in seconds
python scripts/add_identity.py \
    --name "Alice Smith" \
    --folder data/Gallery/aliceGallery \
    --match-tokens alice smith

# 3. (Optional) rebuild full embeddings only if you want alice in train/test splits
python scripts/build_gallery.py
```

3.5 Colab path

Open `notebooks/Colab_Quickstart.ipynb` or `notebooks/Challenge_Day.ipynb` in Colab, run top to bottom. Mounts Drive, sets `FR_DATA_ROOT`, calls the same scripts as above. GPU for free on a T4 or A100 runtime.

3.6 Environment variables

Variable	Effect
`FR_PROJECT_ROOT`	Override repo root (normally auto-detected)
`FR_DATA_ROOT`	Override data path (Colab: `/content/drive/MyDrive/DS_NN_Project_Pictures`)
`FR_ARTIFACTS_DIR`	Where to save `.pkl` / `.pth` outputs

4. In-Class Challenge: Unknown Identity Matching

Rubric §3 · 6% · 4/21 session · 5 unknown feature vectors, 3 correct = full credit

4.1 Input format

The TA distributed 5 `.npy` files, each a dict-wrapped single vector:

- `ID1.npy`, `ID2.npy`, `ID3_UPDATED.npy` — **512-d** ArcFace (InsightFace `buffalo_l`)
- `ID4.npy`, `ID5_UPDATED.npy` — **1000-d** VGG19 **pre-softmax logits** (not the 4096-d fc2 we use by default)

Classmates were photographed in-session (1–3 photos each) and added to Drive as `1Gallery/`… `38Gallery/` folders alongside the 10 original celebrity folders. Total gallery after ingestion: **~180 vectors across 48 identities**.

4.2 Route-by-dimensionality workflow

```
# Sniff dim for each uploaded .npy
import numpy as np
arr = np.load(path, allow_pickle=True)
if arr.dtype == object: arr = arr.item()['features']  # TA's dict wrapper
print(arr.shape)   # (512,) → ArcFace, (1000,) → VGG logits
```

ArcFace unknowns (512-d) — cosine match against the full gallery via `scripts/identify_unknowns.py`:

```
python scripts/identify_unknowns.py /content/ID1.npy --top-k 5
python scripts/identify_unknowns.py /content/ID2.npy --top-k 5
python scripts/identify_unknowns.py /content/ID3_UPDATED.npy --top-k 5
```

VGG unknowns (1000-d pre-softmax logits) — new `scripts/match_vgg_logits.py` extracts matching 1000-d logits from the gallery on-the-fly (VGG19 `fc2 @ W + b` using the `predictions` layer's own weights) and cosine-matches:

```
python scripts/match_vgg_logits.py /content/ID4.npy /content/ID5_UPDATED.npy \
    --classmates-only --center --max-sim --gpu --top-k 5
```

`--classmates-only` drops the 10 celebrity identities so dense celebrity centroids don't drown out sparse classmate centroids. `--center` subtracts the gallery-mean logit (removing the common "face-ish" ImageNet component that squeezes every cosine into 0.70–0.85). `--max-sim` scores each identity by their *closest* gallery image, not the centroid — preserves the one good shot when classmates have only 1–3 photos.

4.3 Source-level patches merged for the challenge

Three bugs surfaced during class and were fixed permanently:

- `walk_image_tasks` previously `break`-ed after the first of `Gallery/` / `Gallery /` was found, silently dropping the other. Now walks both with dedup via a `seen` set.
- `parse_base_identity` had no entry for numeric folder stems, so `1Gallery` … `38Gallery` all parsed to `"Unknown"`. Added a numeric-stem fallback.
- `identify_unknowns.py` couldn't read the TA's object-dtype dict-wrapped `.npy` files (`{'features': array(...)}`). Added a branch that unwraps the dict.

4.4 Results and confidence signals

Unknown	Type	Predicted ID	Similarity	Margin (1→2)	Confidence signal
ID1	ArcFace (512-d)	**22**	0.664	—	High — ArcFace identify() top-1
ID2	ArcFace (512-d)	**8**	0.580	—	High
ID3 (updated)	ArcFace (512-d)	**37**	0.700	—	High
ID4	VGG (1000-d logits)	**24**	0.734 (centroid) / 0.270 (centered+max-sim)	0.034 / 0.017	Moderate — two methods agree
ID5 (updated)	VGG (1000-d logits)	**12**	0.859 (centroid) / 0.408 (centered+max-sim)	0.019 / 0.036	Moderate — two methods agree; #3 drops 25%

Final submission. 5 / 5 predictions produced. The 3 ArcFace matches are high-confidence; the 2 VGG matches are supported by *agreement between two independent scoring schemes* (centroid

on raw logits, and max-similarity on mean-centered logits) — if the features were noise, the two methods would land on different classmates.

Honest caveat on VGG confidence. A k-NN vote over the top-10 nearest gallery images gives only 2/10 agreement for both VGG unknowns — consistent with VGG19's 1000-d ImageNet logits not being face-discriminative at the resolution we need. With 1–3 photos per classmate and 38 classes to discriminate, even a 2/10 vote only barely beats chance (≈1/38 = 2.6/10). The centroid/max-sim agreement is the strongest confidence we can extract from this feature set; better VGG signal would require more photos per classmate, alignment (not just bbox crop), and/or fc2 features instead of pre-softmax logits (all ruled out by the input format the TA provided).

4.5 Where to look in the code

Artifact	Location
ArcFace matcher	`scripts/identify_unknowns.py`
VGG logits matcher	`scripts/match_vgg_logits.py` (flags: `--classmates-only`, `--center`, `--max-sim`, `--knn`)
Challenge notebook	`notebooks/Challenge_Day.ipynb`
Walker + parser patches	`src/feature_extraction.py::walk_image_tasks`, `src/metadata.py::parse_base_identity`

5. Appendix: Full Classifier Comparison

Reproduced verbatim from Stage 3 notebook cell 10 and `scripts/evaluate.py` output. All accuracies on the 20% held-out test set.

Task	Classifier	Overall	Clean	Expression	Lighting	Occlusion	Side
Condition	ArcFace — LR	0.545	0.656	0.462	0.375	0.833	0.500
Condition	VGG19 — LR	**0.612**	0.538	0.500	0.559	0.917	0.857
Identity	ArcFace — LR	**1.000**	1.000	1.000	1.000	1.000	1.000
Identity	VGG19 — LR (raw)	0.810	0.718	0.955	0.882	0.750	0.714
Identity	VGG19 — L2 + PCA + LR	0.785	0.744	0.864	0.824	0.667	0.786

Identity	VGG19 — MLP head	0.818	0.744	0.818	0.912	0.750	0.857
Identity	VGG19 — SupCon + NN centroid	produced by `evaluate.py` step [7]; saved to `artifacts/vgg19_contrastive_head.pth`					
Identity	ArcFace `identify()` cosine	**1.000**	97 / 97 probe images correct				

Deliverables checklist

Rubric item	Where	Status
LR condition classifier on ArcFace/VGG	`src/train.py::train_condition_classifier`	done (0.545 / 0.612)
Condition performance across conditions + failure analysis	§2.1, §2.3 + `evaluate.py` step [9]	done
Train vs test gap analysis	§2.1 last paragraph, §2.3 last paragraph	done
VGG finetune: LR-on-top	`train_vgg_pca_pipeline` (0.785)	done
VGG finetune: layers (MLP head + contrastive)	`train_vgg_mlp_head` (0.818), `train_vgg_contrastive_head`	done
Failure cases for VGG identity	§2.3 per-condition table (Side/Occlusion gap)	done
System documentation	§3 + `README.md`	done
TA reproduction instructions	§3.3 (`build_gallery.py` + `evaluate.py`)	done
5 unknown identity vectors matched	§4.4	5/5 produced
≥3 identities correct (full credit)	§4.4 — 3 ArcFace high-confidence + 2 VGG moderate	targeted